

## BRIEF REPORT

# Factor Structure of the Binge Eating Scale in a Large Sample of Obese and Overweight Patients Attending Low Energy Diet Therapy

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## Abstract

In the present study, we evaluated the dimensionality and psychometric properties of the Italian version of the Binge Eating Scale (BES) in a 669 (127 men and 542 women) obese and overweight patients seeking weight loss treatment. All participants were administered the Italian version of the BES. Fit statistics for the alternative SEM models demonstrated that both the one-factor and competing two-factor models had a comparable fit to the data. Thus, we selected the one-factor model as the most parsimonious. The BES had good internal consistency reliability ( $\alpha = 0.89$ ) with a moderate mean inter-item correlation ( $r_{ii} = 0.34$ ). Given that we were not able to clearly determine the best model among the competing two-factor models and a comparable fit of these models with the one-factor model, we have suggested that the one-factor model is the best fitting model describing the dimensionality of the BES. Copyright © 2015 John Wiley & Sons, Ltd and Eating Disorders Association.

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## Keywords

Binge Eating Scale; obesity; overweight; psychometric properties

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## Introduction

Although the structured clinical interview is considered the gold standard to assess Binge Eating Disorder (BED) (Greeno, Marcus, & Wing, 1995), the Binge Eating Scale (BES) (Gormally, Black, Daston, & Rardin, 1982) is one of the most common instruments used to screen binge eating severity (Marek, Taescavage, Ben-Porath, Ashton, & Heinberg, in press; Ricca et al., 2000). The BES is a 16-item self-report measure developed to assess the behavioural manifestations of binge eating (e.g. eating quickly and overeating) and the feelings/cognitions associated with binge eating (e.g. feeling guilty after binge eating) (Gormally et al., 1982). The BES was found to have good validity and internal consistency ( $\alpha = 0.85$ ) in the original study (Gormally et al., 1982), high two week test-retest reliability ( $r = 0.87$ ), and a significant association with food consumption (Timmerman, 1999). Psychometric properties of the BES were also investigated with satisfactory results in several countries (Dezhkam, Moloodi, Mootabi, & Omidvar, 2009; Freitas, Lopes, Appolinario, & Coutinho, 2006; Ricca et al., 2000; Robert et al., 2013).

Although it was originally developed to assess components of behaviours and feelings/cognitions related to BED, the BES has

been commonly used as a unidimensional measure of binge eating severity. Based on the BES total raw scores, Marcus, Wing, and Hopkins (1988) identified three different levels of severity: individuals scoring 17 or less were considered not reporting significant binge eating, those scoring between 18 and 26 were considered moderate binge eaters, and those scoring 27 and above were considered severe binge eaters. These categories had a 98% concordance rate with a diagnosis using a semi-structured interview (Marcus et al., 1988).

Few studies have investigated the dimensionality of the BES. In the original study, the items were categorized into components of behaviours and feelings/cognitions according to their face validity (Gormally et al., 1982; Timmerman, 1999). More recently, Hood, Grupski, Hall, Ivan, and Corsica (2013) used structural equation modelling with a robust estimator for categorical data to examine the factor structure of the BES in a sample of 530 bariatric surgery candidates. The authors reported that both the unidimensional and the two-factor models provided an adequate fit to the data, although the fit of the two-factor model was better (Hood et al., 2013). Moreover, comparing the fit of the unidimensional and two-factor models in 150 Malay individuals, Robert et al. (2013) reported a better fit

for the two-factor model. Nevertheless, factor composition was different from that suggested in the original study.

Recently, in another sample of bariatric surgery candidates ( $n = 517$ ), Marek, Taescavage, Ben-Porath, Ashton, and Heinberg (2014) compared the one-factor and two-factor models obtained by Hood et al. (2013) with a bi-factor model (i.e. items load contemporaneously on one group factor and a general factor). The authors observed that compared to the one-factor and two-factor models, the bifactor model yielded the best fit to the data. Furthermore, they reported that the feelings/cognitions factor was modestly associated with self-reported mood psychopathology and the behaviour factor was not associated with empirically derived behavioural problems, suggesting poor incremental validity of the behaviour factor (Marek et al., 2014). Thus, although the subfactors of the BES may provide additional information on

binge eating symptoms as well as important treatment indicators (Merrell, 2013), the validity of these subfactors is still unclear.

To the best of our knowledge, no studies have evaluated the factor structure of the BES in a large sample of overweight and obese patients attending low energy diet therapy. Accordingly, the aim of the present study was to investigate the factor structure of the BES by means of confirmatory factor analysis. We compared five factor models reported in previous studies: (i) a one-factor model (Hood et al., 2013); (ii) three two-factor models (Gormally et al., 1982; Hood et al., 2013; Marek et al., 2014; Robert et al., 2013); and (iii) a bi-factor model. We also aimed at investigating the psychometric properties of the best fitting factor model.

## Material and methods

### Participants

Participants were 669 (127 men and 542 women) Italian overweight and obese patients ( $\text{BMI} \geq 25 \text{ kg/m}^2$ ) who were admitted to an outpatient medical centre in Rome ( $n = 497$ ; females 414 and males 83), and the inpatient department of the San Giuseppe Hospital in Verbania (Italy) ( $n = 172$ ; 128 females and 44 males), which specializes in the nutritional treatment of obesity. Clinical and socio-demographic characteristics of the sample are reported in Table 1. The patients were assessed at the time of study entry. None of the patients were bariatric surgery candidates.

The study protocol received ethics approval from the local research ethics review board. Inclusion criteria were ages of 18 and older, and a BMI of 25 or higher. Exclusion criteria were the presence of major disorders of the central nervous system (e.g. epilepsy, dementia, or Parkinson disease), any psychotic symptoms (i.e. hallucinations, delusions, or negative symptoms) or major mood disorders (e.g. bipolar disorder, major depression), purging and non-purging compensatory behaviours, and the presence of any condition affecting the ability to complete the assessment, including the denial of informed consent. After obtaining written informed consent, subjects voluntarily participated in the study and received no payment for their participation.

### Measures

All participants were administered the Italian version of the Binge Eating Scale (BES) (Ricca et al., 2000). Socio-demographic and clinical variables were extracted from medical files. In order to exclude patients with psychotic symptoms and/or major mood

**Table 1** Demographic and clinical data of patients

| Variables                                         | Patients<br>( $n = 497$ ) | Patients<br>( $n = 172$ ) | Total<br>( $n = 669$ ) |
|---------------------------------------------------|---------------------------|---------------------------|------------------------|
| Age— $M$ ( $SD$ )                                 | 44.57 $\pm$ 11.93         | 47.88 $\pm$ 15.43         | 45.42 $\pm$ 12.98      |
| Females— $N$ (%)                                  | 414 (83.3 %)              | 128 (74.4 %)              | 542 (81.0 %)           |
| Males— $N$ (%)                                    | 83 (16.7 %)               | 44 (25.6 %)               | 127 (19.0 %)           |
| School attainment > 13 years— $N$ (%)             | 148 (29.8 %)              | 10 (7.2 %)                | 158 (24.9 %)           |
| Married or in a stable relationship— $N$ (%)      | 313 (63.0 %)              | 82 (59.4 %)               | 395 (62.2 %)           |
| Unemployed— $N$ (%)                               | 107 (23.9 %)              | 58 (42.0 %)               | 165 (24.66 %)          |
| BMI— $M$ ( $SD$ )                                 | 30.37 $\pm$ 5.05          | 44.41 $\pm$ 7.97          | 33.98 $\pm$ 8.54       |
| BMI $\geq 25$ –29.9 $\text{kg/m}^2$ — $N$ (%)     | 288 (57.9 %)              | —                         | 288 (43 %)             |
| BMI between 30 and 34.9 $\text{kg/m}^2$ — $N$ (%) | 135 (27.2 %)              | 14 (8.1 %)                | 149 (22.3 %)           |
| BMI between 35 and 39.9 $\text{kg/m}^2$ — $N$ (%) | 51 (10.3 %)               | 40 (23.3 %)               | 91 (13.6 %)            |
| BMI $\geq 40 \text{ kg/m}^2$ — $N$ (%)            | 23 (4.6 %)                | 118 (68.6 %)              | 141 (21.1 %)           |
| BES— $M$ ( $SD$ )                                 | 13.35 $\pm$ 8.93          | 15.35 $\pm$ 10.47         | 13.86 $\pm$ 9.40       |
| BES $\leq 17$ — $N$ (%)                           | 349 (70.2 %)              | 108 (62.8 %)              | 457 (68.3 %)           |
| BES between 18 and 26— $N$ (%)                    | 107 (21.5 %)              | 37 (21.5 %)               | 144 (21.5 %)           |
| BES $\geq 27$ — $N$ (%)                           | 41 (8.2 %)                | 27 (15.7 %)               | 68 (10.2 %)            |

Note:

BMI = body mass index; BES = Binge Eating Scale.

**Table 2** Fit statistics for the alternative models (estimator: robust diagonally weighted least squares)

|                                                             | Satorra–Bentler<br>Scaled $\chi^2$ | Root Mean square<br>error of approximation<br>(RMSEA) (90%CI) | Comparative<br>fit index<br>(CFI) | Standardized root<br>mean square<br>residual (SRMSR) | Degree of<br>freedom<br>(df) | Akaike<br>information<br>criterion (AIC) | Expected<br>cross-validation<br>index (ECVI) |
|-------------------------------------------------------------|------------------------------------|---------------------------------------------------------------|-----------------------------------|------------------------------------------------------|------------------------------|------------------------------------------|----------------------------------------------|
| One-factor model                                            | 302.45*                            | 0.053 (0.046/0.061)                                           | 0.99                              | 0.053                                                | 104                          | 366.45                                   | 0.55 (0.48/0.63)                             |
| Two-factor model (Gormally et al., 1982)                    | 271.67*                            | 0.050 (0.042/0.057)                                           | 0.99                              | 0.050                                                | 103                          | 337.67                                   | 0.51 (0.44/0.58)                             |
| Two-factor model (Hood et al., 2013;<br>Marek et al., 2014) | 263.00*                            | 0.048 (0.041/0.055)                                           | 0.99                              | 0.049                                                | 103                          | 329.00                                   | 0.49 (0.43/0.57)                             |
| Two-factor model (Robert et al., 2013)                      | 264.78*                            | 0.048 (0.041/0.056)                                           | 0.99                              | 0.050                                                | 103                          | 330.78                                   | 0.50 (0.43/0.57)                             |
| Bi-factor model (Hood et al., 2013;<br>Marek et al., 2014)  | 152.50*                            | 0.033 (0.024/0.042)                                           | 1.00                              | 0.036                                                | 88                           | 248.50                                   | 0.37 (0.33/0.43)                             |

\*Significant for  $p < 0.001$ .

disorders, at the study entry, participants were also asked screening questions according to a checklist (Innamorati et al., 2014) based on the *Diagnostic and Statistical Manual of Mental Disorders 4th edition, text revision* (DSM-IV-TR) (American Psychiatric Association, 2000) criteria for psychotic and major mood disorders.

### Statistical analyses

Confirmatory factor analysis was used to compare five factor models: (i) a one-factor model with all items loading on the latent factor; (ii) a two-factor (feelings/cognitions and behaviours) model described by Gormally et al. (1982) (feelings/cognitions items: 1-3-6-7-12-14-15-16; behaviours items: 2-4-5-8-9-10-11-13); (iii) a two-factor model reported in bariatric surgery candidates (Hood et al., 2013; Marek et al., 2014) (feelings/cognitions items: 1-3-6-7-12-14-15; behaviours items: 2-4-5-8-9-10-11-13-16); (iv) a two-factor model reported in Robert et al. (2013) (feelings/cognitions items: 1-2-6-9-11-12-14-16; behaviours items: 3-4-5-7-8-10-13-15); and (v) a bi-factor model reported in Marek et al. (2014), where the composition of the groups factors (feelings/cognitions and behaviours) was the same as reported by Hood et al. (2013). In a bi-factor model, the group factors and the general factor were uncorrelated.

All analyses were performed using the statistical package Lisrel 8.8 (Jöreskog & Sörbom, 2006). We used a Robust Diagonally Weighted Least Squares estimator (DWLS) on a polychoric correlational matrix after having calculated Mardia's (1970) coefficient for multivariate kurtosis. Although maximum likelihood and generalized least squares are commonly used in confirmatory factor analysis, they assume that the data follow a multivariate normal distribution, whereas, DWLS is more robust to deviations from these assumptions (Flora & Curran, 2004). We reported the following model fit indices: (i) The Satorra-Bentler scaled chi-square ( $\chi^2$ ). Non-significant values indicate good model fit. Nevertheless, this index is dependent on sample size, and tends to be significant in large samples (Bentler & Bonett, 1980); (ii) The Root Mean Square Error of Approximation (RMSEA). Values between 0.05 and 0.08 indicate an acceptable fit of the model tested; whereas, values lower than 0.05 are indicative of good fit (Browne & Cudek, 1993), while Hu and Bentler (1999) recommended values lower than 0.06; (iii) The Incremental Comparative Fit Index (CFI). Values higher than 0.95/0.96 indicate a good model fit (Hu & Bentler, 1999); (iv) The Standardized Root Mean Square Residual (SRMR). Values close to zero are indicators of perfect fit of the model. Values lower than 0.08 are indicators of satisfactory model fit (Hu & Bentler, 1999). Models were compared by means of the Expected Cross-Validation Index (ECVI) (Browne & Cudek, 1993), and the Akaike's Information Criterion (Akaike, 1973). The ECVI is the average discrepancy in the fitted covariance matrices between two samples of equal sample size across all possible combinations of two samples from the same population (Cudek & Browne, 1983). The model having the smallest ECVI value exhibits the greatest potential for replication.

As measures of reliability, we reported McDonald's (1999) omega (the proportion of total score variance that can be attributed to all factors, group and general factors), omega *H* (the proportion of total score variance attributable to the general factor), and omega *S* (the proportion of total score variance that can be attributed to all group factors, after controlling for the general

**Table 3** Standardized factor loadings ( $R^2$ ) for the alternative models ( $N = 669$ )

| Items            | One-factor model    |             | Two-factor model (Gormally et al., 1982) |             | Two-factor model (Hood et al., 2013; Marek et al., 2014) |             | Two-factor model (Robert et al., 2013) |             | Bi-factor model (Marek et al., in Press)                            |            |                 |
|------------------|---------------------|-------------|------------------------------------------|-------------|----------------------------------------------------------|-------------|----------------------------------------|-------------|---------------------------------------------------------------------|------------|-----------------|
|                  | Feelings/cognitions | Behaviours  | Feelings/cognitions                      | Behaviours  | Feelings/cognitions                                      | Behaviours  | Feelings/cognitions                    | Behaviours  | Feelings/cognitions                                                 | Behaviours | General factors |
| Item 1           | 0.60 (0.35)         | —           | 0.60 (0.36)                              | —           | 0.61 (0.37)                                              | —           | 0.61 (0.37)                            | —           | 0.55                                                                | —          | 0.49 (0.55)     |
| Item 2           | 0.41 (0.17)         | 0.42 (0.18) | —                                        | 0.42 (0.18) | —                                                        | 0.42 (0.18) | 0.42 (0.18)                            | —           | —                                                                   | 0.02       | 0.42 (0.18)     |
| Item 3           | 0.73 (0.53)         | —           | 0.74 (0.54)                              | —           | 0.74 (0.55)                                              | —           | —                                      | 0.74 (0.54) | 0.12                                                                | —          | 0.72 (0.53)     |
| Item 4           | 0.81 (0.66)         | 0.83 (0.68) | —                                        | 0.82 (0.68) | —                                                        | 0.82 (0.68) | —                                      | 0.82 (0.68) | —                                                                   | —          | 0.84 (0.78)     |
| Item 5           | 0.61 (0.38)         | 0.62 (0.39) | —                                        | 0.62 (0.39) | —                                                        | 0.62 (0.39) | —                                      | 0.62 (0.39) | —                                                                   | 0.16       | 0.62 (0.41)     |
| Item 6           | 0.69 (0.48)         | —           | 0.70 (0.49)                              | —           | 0.71 (0.50)                                              | —           | —                                      | —           | 0.53                                                                | —          | 0.60 (0.64)     |
| Item 7           | 0.68 (0.46)         | —           | 0.69 (0.48)                              | —           | 0.69 (0.48)                                              | —           | —                                      | 0.69 (0.48) | 0.10                                                                | —          | 0.67 (0.46)     |
| Item 8           | 0.64 (0.41)         | 0.65 (0.42) | —                                        | 0.65 (0.43) | —                                                        | 0.65 (0.43) | —                                      | 0.65 (0.42) | —                                                                   | 0.07       | 0.65 (0.43)     |
| Item 9           | 0.55 (0.30)         | 0.56 (0.32) | —                                        | 0.56 (0.32) | —                                                        | 0.56 (0.32) | —                                      | —           | —                                                                   | —          | 0.56 (0.32)     |
| Item 10          | 0.79 (0.62)         | 0.80 (0.65) | —                                        | 0.81 (0.65) | —                                                        | 0.81 (0.65) | —                                      | —           | —                                                                   | 0.01       | 0.81 (0.65)     |
| Item 11          | 0.81 (0.65)         | 0.82 (0.68) | —                                        | 0.83 (0.68) | —                                                        | 0.83 (0.68) | —                                      | —           | —                                                                   | 0.27       | 0.82 (0.74)     |
| Item 12          | 0.68 (0.46)         | —           | 0.69 (0.47)                              | —           | 0.69 (0.48)                                              | —           | —                                      | —           | 0.35                                                                | —          | 0.62 (0.51)     |
| Item 13          | 0.67 (0.45)         | 0.68 (0.47) | —                                        | 0.68 (0.47) | —                                                        | 0.68 (0.47) | —                                      | 0.68 (0.46) | —                                                                   | —          | 0.70 (0.54)     |
| Item 14          | 0.74 (0.55)         | —           | 0.75 (0.56)                              | —           | 0.76 (0.57)                                              | —           | —                                      | —           | 0.37                                                                | —          | 0.68 (0.60)     |
| Item 15          | 0.75 (0.56)         | —           | 0.76 (0.57)                              | —           | 0.76 (0.58)                                              | —           | —                                      | 0.76 (0.57) | 0.05                                                                | —          | 0.75 (0.56)     |
| Item 16          | 0.57 (0.33)         | —           | 0.58 (0.34)                              | —           | —                                                        | 0.59 (0.34) | —                                      | 0.59 (0.34) | —                                                                   | 0.21       | 0.58 (0.38)     |
| McDonald's Omega | 0.61                | —           | 0.46                                     | —           | 0.46                                                     | —           | 0.47                                   | —           | Omega = 0.66; $H = 0.63$ ; $S_{reel} = 0.12$ ; $S_{behav} = 0.0008$ |            |                 |

All factor loadings are significant for  $p < 0.05$ .

One-factor model statistics (general factor): Cronbach  $\alpha = 0.89$ ; mean inter item correlation = 0.34.

factor) indices hand calculated according the procedure suggested in Reise, Bonifay, and Haviland (2013). Pearson's  $r$  indices of correlations and  $t$ -tests were used to examine the association among BES scores and sociodemographic variables.

## Results

### Model fit and internal consistency

Mardia's (1970) coefficient for multivariate kurtosis (334.536;  $df=816$ ;  $p < 0.001$ ) was significant, indicating severe non-normal distribution of the variables and suggesting the use of the DWLS estimator on a polychoric correlational matrix. Fit statistics for the alternative SEM models are reported in Table 2, with factor loadings presented in Table 3. The analyses demonstrated that all the models had an adequate fit to the data, despite a significant  $\chi^2$ . When comparing the models, the bi-factor model had the lowest AIC and ECVI, denoting superiority in its potential for replication when compared with the one-factor and the two-factor models.

The bi-factor model also had the highest McDonald's Omega coefficient (0.66), while the two-factor models had the lowest Omega coefficients (0.46–0.47) (Table 3), indicating that the bi-factor model may explain a greater percentage of the variance in observed scores. However, McDonald's Omega H and S coefficients demonstrated that only the general factor explained a significant percentage of variance in observed scores, whereas, the variance explained by the behaviour factor is negligible. Furthermore, almost all items loaded more strongly on the general factor than the corresponding group factor, with the exception of item 1, which loaded more strongly on the feelings/cognitions factor, indicating quasi-unidimensionality.

Small correlations were found among the BES, BMI ( $r = -0.14$ ;  $p < 0.001$ ), and the age of the participants ( $r = -0.14$ ;  $p < 0.001$ ). Compared to male patients, female patients reported significantly higher BES scores ( $9.98 \pm 8.25$  vs.  $14.77 \pm 9.40$ ;  $t_{209,73} = -5.74$ ;  $p < 0.001$ ).

## Discussion

Our results indicated that although all of the models compared had an adequate fit to the data, the bi-factor model was more parsimonious and superior in its potential for replication. However, reliability coefficients and the pattern of factor loadings suggested a quasi-unidimensionality of the BES. Thus, our results cast doubt on the utility of calculating the behaviours and feelings/cognitions factors as separate scores. This is consistent with results of Marek et al. (2014) who reported that, although the two-factor model yielded a good fit, the 2 factors were strongly related ( $r = .89$ ) and that the behaviour factor was characterized by poor

incremental validity. The general factor also had moderate mean inter-item correlation ( $r_{ii} = 0.34$ ) comparable to the results of Robert et al. (2013) in Malay individuals.

Consistent with previous studies (Dalle Grave, Calugi, Petroni, Di Domizio, & Marchesini, 2010; Delahanty, Meigs, Hayden, Williamson, & Nathan, 2002; Ricca et al., 2000), we also found a small, but significant correlation between the BES and the BMI. However, inconsistent with the results of Ricca et al. (2000) who reported a small but significant positive association ( $r = 0.12$ ), we observed a small, but negative correlation between the BES and age.

Finally, in line with previous findings (Presnell, Pells, Stout, & Musante, 2008; Ricca et al., 2000), compared to men, women reported higher BES total scores. According to recent neuroimaging studies (Geliebter et al., 2013; Wang et al., 2009), sex differences in binge eating may reflect the presence of some alteration in the female brain's reward system, such as the lower ability to suppress hunger in women (Wang et al., 2009).

There are some limitations in generalizing the results. First, our sample included mostly slightly obese female patients; thus, results may be different in samples of patients with severe obesity and among male patients. Second, we did not administer measures of depression or anxiety. Although we excluded patients with psychotic symptoms and/or major mood disorders, sub-threshold depression or anxiety may have been present and are known to be associated with BES scores (Ricca et al., 2000). Third, we did not assess the presence of eating disorders or dysfunctional eating pattern such as food addiction or night eating syndrome which are known to be associated with BES (Harb et al., 2012; Imperatori et al., 2014). Finally, test–retest stability was not investigated. On the other hand, to our knowledge, this is the first study examining the factor structure of the BES in an Italian sample of obese and overweight patients. Last, we did not directly investigate incremental validity of the subfactors' scores.

In conclusion, our results suggest that the Italian version of the BES is nearly unidimensional and the BES total score is a reliable measure of binge eating severity. Although some authors have hypothesized that considering the feelings/cognitions and behaviour factors may provide clinicians with additional information on binge eating symptoms as well as important treatment indicators, our results revealed poor reliability of the subfactors, suggesting that calculating these scores may be not useful.

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